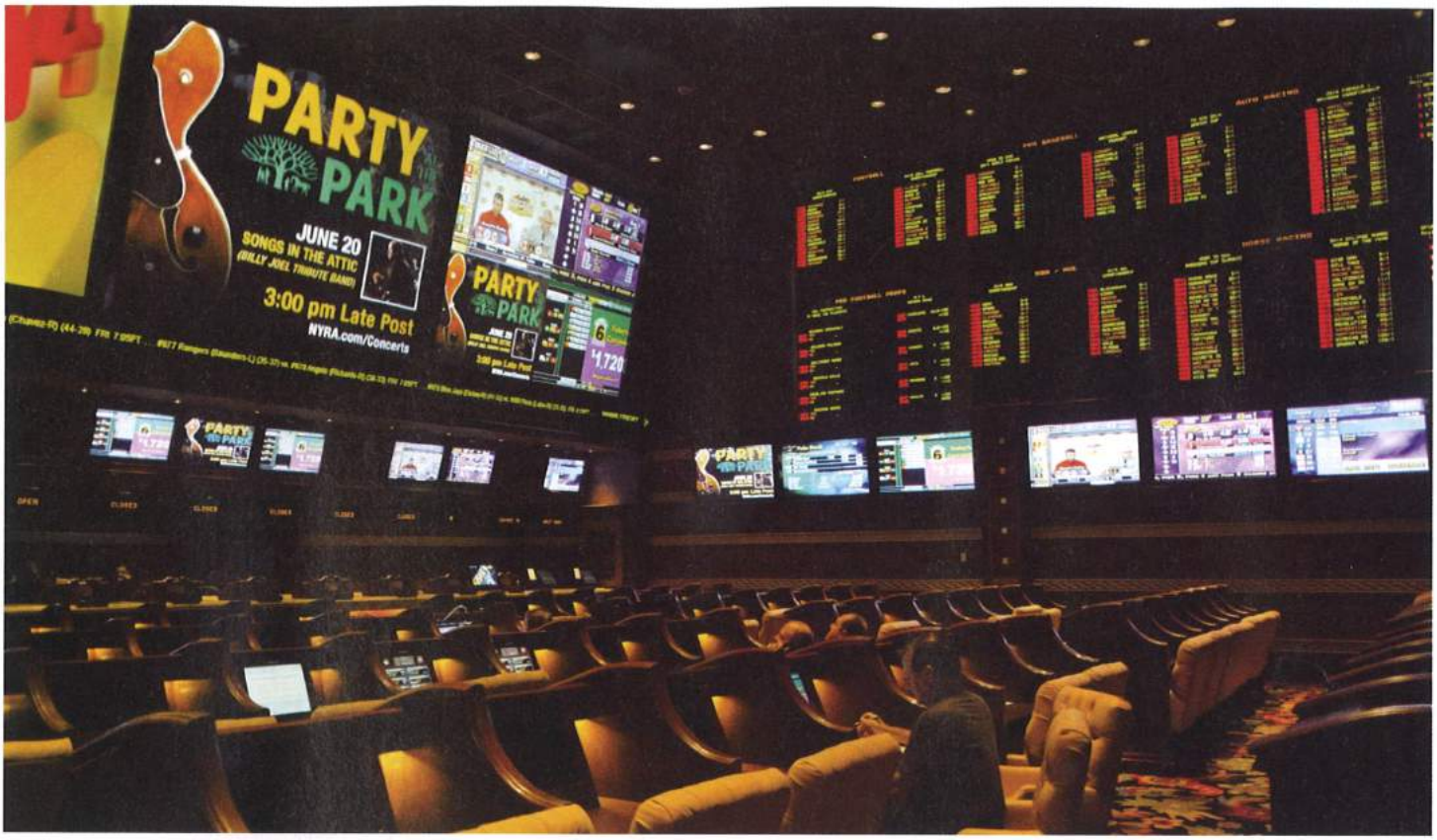


The SVSI system by Harman AMX at Wynn Resort requires less rack space than a traditional matrix switcher by using standard IT network hardware

Are you sending all the right signals?

Signal delivery, switching and processing can be a complex area to master. *Paul Bray* talks to experts in their field to help explain the processes and terminology.

Most of us are familiar with the inputs and outputs of an AV system – the computers, media players or cameras and microphones that provide the content, and the screens, projectors and speakers on which we enjoy or endure it. But what happens in the middle? Is there a straight highway from input to output that we can navigate for ourselves, or a Spaghetti Junction of baffling complexity that's best left to experts?

It all depends, says Joseph Barbier, marketing manager of digital connectivity vendor, DVIGear. "In a simple workstation application, a single HDMI cable may be all you need to transport the content from PC to monitor. But larger installations such as boardrooms, command and control centres or digital signage have challenges that require additional equipment to perform signal switching, processing and delivery."

AV like mamma used to make

Unravel the spaghetti and you'll probably end up with three strands: multiple sources and displays, incompatible systems and connectors, and long distances.

"You rarely see a true 'one source/one display' set-up," says Kyle Davis, product marketing manager at control systems vendor, AMX. "So you need a way to switch video between those sources and displays, whether that's a physical box (called a video switcher) or virtualised switching and delivery over the IT network (streaming video).

"A second cause of complexity is when the source device isn't using the

Integrator Vidéosonic, in collaboration with manufacturer Analog Way and software development company Technidream, created a tailor-made installation at French banking group, the Fédération Nationale du Crédit Agricole duCrédiAgricol(FNCA)

same connection type as the display. You can't plug a VGA cable into an HDMI port, for example. There are many options, from simple adaptor cables (which let the devices physically connect, but don't change the 'language' the cable is speaking) to processing and translating that signal into one that the display understands.

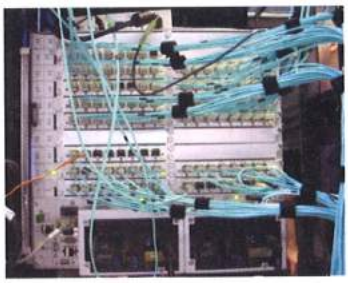
"Finally, each type of cable has its own distance limitation; with HDMI in particular this can be fairly short and it's where technologies such as HDBaseT, fibre and networked AV come into play. They let you send video hundreds or thousands of feet – or in the case of networked AV, around the globe."

Take a command and control environment, says Barbier. It may have many operators at small workstations plus some large-scale videowalls or projection systems. This requires specialised equipment to manage and route all the video signals from their points of origin to the desired locations and switch them to different output devices. The sources and displays may also be far apart, so the signal needs to be boosted over distance. Add other technical challenges such as copy protection and multiple display resolutions and it's time to start ladling the Bolognese sauce.

Switching, processing and delivery

The technologies that ravel up our strands into dear old Spaghetti Junction again can be roughly characterised as signal switching, processing and delivery. Sometimes these functions are incorporated into the video sources and displays, but they are often standalone devices that give a greater range of functionality and control.

Switchers are selectors that enable users to choose between several AV sources."Think of AV sources as tv channels, the switch as the tv cable box and the remote as the tool to manage the switch," says Samuel Recine, director of sales, Americas and Asia Pacific at vendor, Matrox Graphics. "The switch is something that lets you change the channel on the viewing station."



tvONE Voyager Matrix switch at Greene County Office of Emergency Management

Delivery comprises two different functions: extending and distributing, says Stephan Vinke, a field trainer at signal management and distribution specialist, Atlona. "Extenders allow a signal to travel longer distances than the native signal permits. Distributors enable (multiple) delivery by accepting one input signal and allocating it to several outputs," he says.

Simple switching and delivery don't change the properties of the AV signal, such as resolution, refresh rate or aspect ratio. So when the source and output have different interfaces or capabilities you will probably also need processing technology.

"**Processors** include several types of device," says Vinke. "Converters change the electrical interface of AV signals (eg. from analogue VGA to digital HDMI). Scalers increase or decrease the resolution. Audio processors decode audio signals to create a surround sound environment and can insert audio into an HDMI signal (embedding) or extract audio from it (de-embedding)."

Switching and scaling are sometimes combined in the same device, while so-called seamless switchers can switch sources without the screen going momentarily blank in between, according to Franck Facon, marketing and communications director at signal processing and distribution vendor, Analog Way.

"Extension is needed whenever your sources and displays are not in the same room or are too far apart for a standard HDMI/VGA/DP cable," explains Mark Armon, product manager at video conversion and signal distribution vendor, tvONE. "Switching lets you see the same information in multiple places or change what's available to be seen in different places. Processing is only required if you're changing the original content into something else, such as a videowall, or for compressing a signal to fit on to an older infrastructure."

Key ingredients

The selection of particular technologies may be dictated by specific needs.

"It's possible to optimise for **latency** (eg. for events where live action and recorded material must be perfectly synchronised)," says Recine. "This would probably favour technologies that move well over distance in an uncompressed fashion and can sustain quality, all at low latency, such as SDI extension/switching or HDBaseT.

"It's also possible to optimise for **scalability**, both many-to-many (many sources to many consumers) and any-to-any (camera, PC, microphone to tv, videowall, phone and tablet). The unequivocal choice here is **AV-over-IP**. Once the source is an IP stream it can be cost-effectively extended, switched over standard IP switches, stored compressed, transmitted by wire or wirelessly (eg. to smartphones and tablets), and more."

IP conversion requires an **encoder** or transmitter (already included in certain devices, such as professional cameras). In essence this converts AV content into computer data which can then be switched, processed, distributed and delivered to just about any networked device.

"(Encoders) ingest all required AV signal types in their native or raw formats (eg. TMDS for HDMI), and package them together such that all these signals can be transmitted over a single cable at much longer distances,"

LEARNING THE LINGO

HDMI is the most common and flexible standard for conveying video and audio, well established and widely supported. But its range is limited, so a newer standard, **HDBaseT** is also popular for realtime, uncompressed transmissions, giving a range of up to 100 metres or so. **SDI** (good for realtime video), **DisplayPort** (with its high resolution) and **DVI** are quite common alternatives.

Commercial content is likely to be copy-protected using **HDCP** (High-bandwidth Digital Content Protection). HDMI, DVI and DisplayPort all carry it, but some displays and projectors aren't compatible and won't display protected content, while other output devices may have HDCP activated by default and fail to display certain content. **HDCP management** is therefore desirable in signal distribution devices.

AVB (audio video bridging) allows the routing of audio and video over off-the-shelf IP switches in a synchronous, reliable and predictable way.

Crestron's DM-RMC-4K- SCALER-C 4K control interface also functions as a receiver and video scaler. The HD-SCALER-HD-E enables HD, SD or computer display devices to handle any input resolution up to Full HD 1080p or WUXGA

says Kamran Ahmed, ceo of signal processing and distribution vendor, AptoVision.

"Look for encoders that can produce quality, low (and adjustable) bit rates, high resolution (4K encoders are already becoming available) and usability (management software)," advises Recine.

At the other end is a **decoder** or receiver. "This is connected to the display device, and it receives and unpacks the various signals and restores them to their original formats," says Kamran. "In an uncompressed signal distribution system it's critical that this doesn't introduce any pixel errors or significant delays (ie. beyond microseconds).

"To enable distribution of the AV signal across an AV network, a **matrix switching system** (in the case of HDBaseT) or an off-the-shelf IP switch is needed between the AV transmitter and receiver. This routes the signals based on commands from a control system such that a source can be simultaneously broadcast to one or more displays or a display can receive AV signals from any source."

"There's often a further need for up-down-cross conversion to alter raster sizes, and for a frame synchroniser to stabilise the signal and correct flicker," adds Bryce Button, product marketing manager at digital video vendor, AJA. "If the source might be coming off a computer then it helps to have a product that allows you to pull a region of interest in real-time from your frame or desktop for broadcast. Sometimes you need to convey signals over local IT infrastructure cables such as Cat 5 or Cat 6, when products like HDBaseT mini-converters will be essential."

The pasta chef

Can you cook up these ingredients for yourself, or should you hire a voluble Italian in a tall, white hat? Again it depends, says Vinke. "If the system is simple and has a 1:1 aspect ratio, you just need to check if sources and outputs both support the same AV interface (HDMI, DVI, DisplayPort, HDBaseT), and if the distance is no more than five metres a connecting cable is all you need.

"However, if the distance is large, interfaces are different, several sources need to be shown on one display, or a signal needs to be shown at the same time on different displays, then an integrator can certainly help."

Finally, warns Armon, don't order a feast if you're not very hungry. "Be clear from the outset what level of quality you're looking for," he says. "For example, does a compressed signal look 'good enough'? Does latency matter? Do you need all your screens to be exactly in sync? It's entirely possible to deliver outstanding, uncompressed video content to every screen, with zero latency and perfect synchronisation. But there will be a price." ■ »

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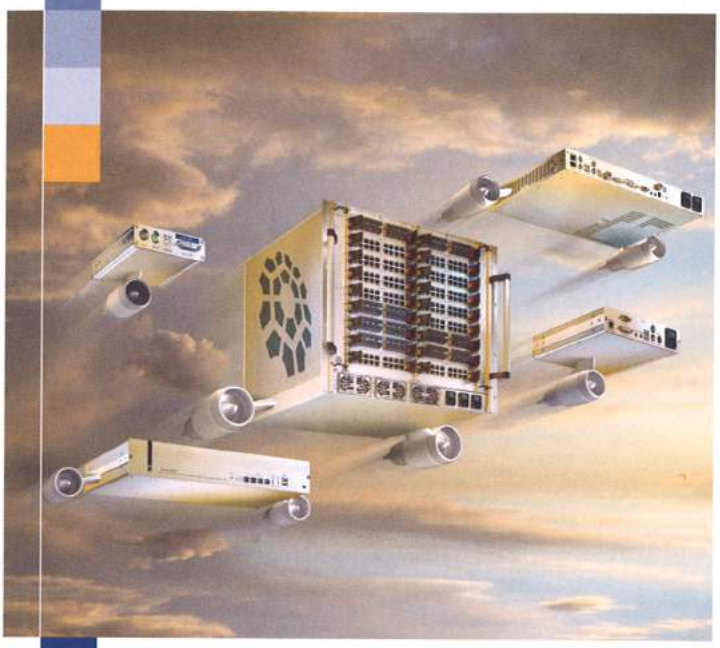
IT'S WHAT'S BEHIND THAT KEEPS US AHEAD



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